

| <b>PROJECT</b>                                          |                 | <b>JOB NO.</b>                                                                    |                                                    |    |    |         |
|---------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------|----------------------------------------------------|----|----|---------|
| <b>CONTRACTOR :</b>                                     |                 |                                                                                   |                                                    |    |    |         |
| <b>SUB-CONTRACTOR : Winner Engineering Pte Ltd</b>      |                 |                                                                                   |                                                    |    |    |         |
| <b>TRADE</b>                                            | : ACMV SERVICES | <b>DRAWING NO :</b>                                                               |                                                    |    |    |         |
| <b>PRE - TESTING COMMISSIONING CHECKLIST ( MV FAN )</b> |                 |                                                                                   |                                                    |    |    |         |
| <b>Unit Designation :</b>                               |                 |                                                                                   |                                                    |    |    |         |
| <b>Unit Make :</b>                                      |                 |                                                                                   |                                                    |    |    |         |
| <b>Model no.</b>                                        |                 |                                                                                   |                                                    |    |    |         |
| <b>Location :</b>                                       |                 |                                                                                   |                                                    |    |    |         |
| <b>Serial no :</b>                                      |                 |                                                                                   |                                                    |    |    |         |
| S/N                                                     | Section         | Inspection / Test item                                                            | CHECKED CONDITION<br>(Please tick inside checkbox) |    |    | Remarks |
|                                                         |                 |                                                                                   | Yes                                                | No | NA |         |
| 1                                                       | Documents       | Verify fan tag, manufacturer, model, serial number and motor nameplate.           |                                                    |    |    |         |
| 2                                                       | Documents       | Verify fan, grille, louver, damper and duct layout against approved drawings.     |                                                    |    |    |         |
| 3                                                       | Documents       | Verify electrical schematic and approved control sequence are available.          |                                                    |    |    |         |
| 4                                                       | Access          | Check maintenance access to fan, motor, belts, bearings, dampers and isolator.    |                                                    |    |    |         |
| 5                                                       | Installation    | Check cleanliness and condition of fan, housing and nearby area.                  |                                                    |    |    |         |
| 6                                                       | Installation    | Check fan base, mounting, supports, fasteners and alignment.                      |                                                    |    |    |         |
| 7                                                       | Installation    | Check anti-vibration isolators and removal of transport restraints.               |                                                    |    |    |         |
| 8                                                       | Mechanical      | Check impeller, blades, shaft and moving parts with power isolated.               |                                                    |    |    |         |
| 9                                                       | Mechanical      | Check fan and motor bearings and lubrication where required.                      |                                                    |    |    |         |
| 10                                                      | Mechanical      | Check belt condition, pulley alignment and belt tension if belt-driven.           |                                                    |    |    |         |
| 11                                                      | Mechanical      | Check coupling/alignment if coupling-driven and guards over exposed moving parts. |                                                    |    |    |         |
| 12                                                      | Ductwork        | Check ductwork completion, supports and access panels.                            |                                                    |    |    |         |
| 13                                                      | Ductwork        | Check flexible connections at fan inlet/outlet where provided.                    |                                                    |    |    |         |
| 14                                                      | Ductwork        | Check duct joints, seals and specified duct leakage-test records.                 |                                                    |    |    |         |
| 15                                                      | Ductwork        | Verify fan, duct, airflow-direction and circuit identification labels.            |                                                    |    |    |         |
| 16                                                      | Air path        | Check grilles, diffusers, louvers, screens and fan inlet/outlet.                  |                                                    |    |    |         |
| 17                                                      | Air path        | Check filters where specified and confirm temporary-filter arrangements.          |                                                    |    |    |         |
| 18                                                      | Dampers         | Check VCD installation, access and initial commissioning position.                |                                                    |    |    |         |
| 19                                                      | Dampers         | Check fire/smoke dampers and access where provided.                               |                                                    |    |    |         |

|    |                   |                                                                                                  |  |  |  |  |
|----|-------------------|--------------------------------------------------------------------------------------------------|--|--|--|--|
| 20 | Dampers           | Check motorized damper actuators, linkages and wiring where fitted.                              |  |  |  |  |
| 21 | Dampers           | Check non-return damper installation and blade clearance where fitted.                           |  |  |  |  |
| 22 | Terminals         | Check VAV boxes or controlled terminals where connected.                                         |  |  |  |  |
| 23 | Electrical        | Verify LCP/LMCP labels and breaker/contactors ratings against single-line diagram.               |  |  |  |  |
| 24 | Electrical        | Check cable condition, termination tightness, glands and circuit labels with power isolated.     |  |  |  |  |
| 25 | Electrical        | Verify earthing continuity and cable/motor insulation-resistance test records.                   |  |  |  |  |
| 26 | Electrical        | Check overload/protection settings and applicable protective-device test records.                |  |  |  |  |
| 27 | VSD               | Verify VSD motor data, rated current/kW, speed limits and bypass wiring where installed.         |  |  |  |  |
| 28 | Controls          | Check BMS wiring and installed sensors, switches and feedback devices.                           |  |  |  |  |
| 29 | Car-park controls | Verify CO sensor location, calibration records and associated fan/zone mapping where applicable. |  |  |  |  |
| 30 | Readiness         | Verify test instrument calibration, test procedure and outstanding pre-start defects.            |  |  |  |  |
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Comments to follow up Action :

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Yes = satisfactory; No = rectify and re-inspect; NA = not applicable (state reason). Blank = not checked.

| CHECKED BY:   |  | WITNESSED BY | INSPECTED BY |
|---------------|--|--------------|--------------|
| NAME          |  |              |              |
| DESIGNATION : |  |              |              |
| COMPANY :     |  |              |              |
| SIGNATURE :   |  |              |              |
| DATE :        |  |              |              |